

Dirichlet-Multinomial Error-Choice Simulation ($p_{\min} = 0.01$)

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1 Overview

This report evaluates **dirichlet-multinomial** sampling error across cell-type proportions generated from a weighted $\text{Beta}(\alpha, 1)$ distribution, with the smallest population proportion pinned at $p_{\min} = 0.01$.

1.1 Success rule

For each replicate and cell type j , the estimated proportions are averaged over the N people in that replicate. The average is compared with the population-level true proportion p_j :

$$\bar{p}_j = \frac{1}{N} \sum_{i=1}^N \hat{p}_{ij}, \quad AE_j = |\bar{p}_j - p_j|, \quad ARE_j = \frac{|\bar{p}_j - p_j|}{p_j}. \quad (1)$$

A replicate **succeeds** for AE if $\max AE_j \leq \tau_{AE}$, and for ARE if $\max ARE_j \leq \tau_{ARE}$. AE and ARE are always evaluated separately and never combined into one criterion.

2 Simulation setup

The data are generated in two steps (a Dirichlet-multinomial hierarchy):

1. Person i has a true composition $\theta_i \sim \text{Dirichlet}(c \cdot p)$, where p is the population mean composition and c is the **concentration**.
2. From that person, n cells are sampled: $\text{counts}_i \sim \text{Multinomial}(n, \theta_i)$, so $\hat{p}_{ij} = \text{count}_{ij}/n$.

The population proportions p are fixed values derived from a Beta distribution, with the smallest proportion pinned (Table 2, Figure 1). The construction starts from the plain $\text{Beta}(\alpha, 1)$ proportions on a grid of K points. The smallest type, and every type that would fall below $p_{\min}$, is set to exactly $p_{\min}$. The remaining types are rescaled proportionally so that all proportions still sum to one. A larger α makes the composition more skewed, so more types are set to $p_{\min}$: 1, 2, 3, 4 types for $\alpha = 2, 3, 4, 5$ respectively. Each scenario (α, N, n) is simulated B times. Table 1 lists all settings.

Table 1: Simulation settings (`simulation_dm_errorchoice_pmin_defaults()`).

Parameter	Value
alpha (Beta shape)	2, 3, 4, 5
K (cell types)	10
B (replicates per scenario)	1,000
Concentration c	10000
Proportion method	fixed_min_beta
p_min	0.01
N (people per replicate)	1, 2, 3, 5, 10
n per person, sweep	100 to 100,000,000 (25 log-spaced values)
n per person, threshold sweep	200,000
tau_AE (fixed)	0.005
tau_ARE (fixed)	0.5
Target success rate	0.95
Seed	260926

Table 2: Population mean proportions p_j per cell type (columns) for each α (rows). The smallest type and every type that would fall below $p_{\min} = 0.01$ are set to exactly 0.01, so 1, 2, 3, 4 types are set to $p_{\min}$ for $\alpha = 2, 3, 4, 5$ respectively.

alpha	1	2	3	4	5	6	7	8	9	10
2	0.01	0.03	0.05	0.07	0.09	0.11	0.13	0.15	0.17	0.19
3	0.01	0.01	0.019	0.036	0.06	0.09	0.13	0.17	0.21	0.27
4	0.01	0.01	0.01	0.017	0.036	0.065	0.11	0.17	0.24	0.34
5	0.01	0.01	0.01	0.01	0.02	0.045	0.087	0.15	0.26	0.4

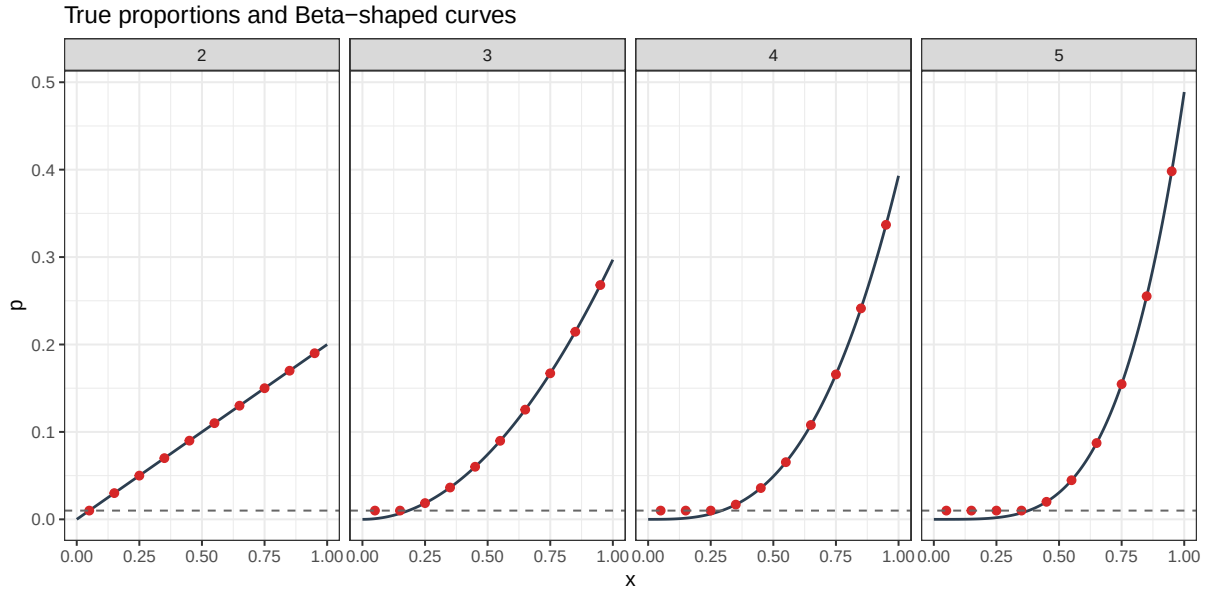


Figure 1: Construction of the population proportions: points are the proportions p_j per cell type, the curve is the underlying scaled Beta curve, and the dashed line marks $p_{\min} = 0.01$. One panel per α .

3 Simulation results

3.1 Success rate versus threshold

Figure 2 and Figure 3 show the success rate as a function of the threshold τ at a fixed 200,000 cells per person.

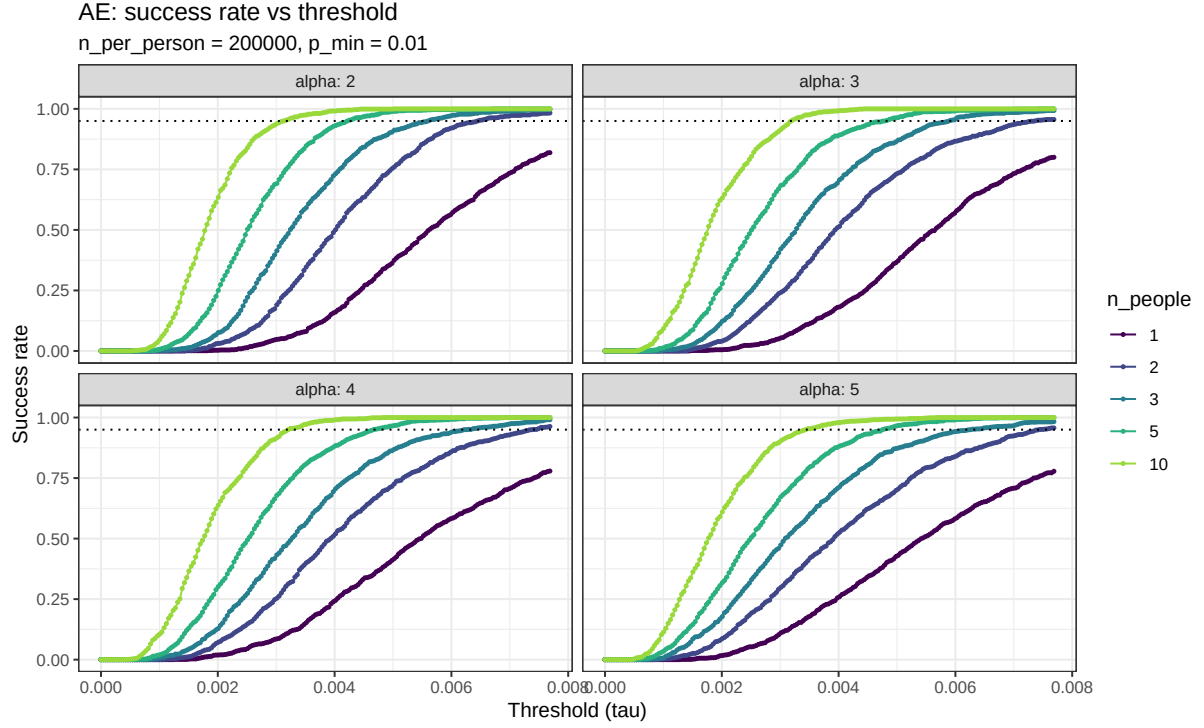


Figure 2: AE success rate versus threshold τ_{AE} at 200,000 cells per person. One panel per α , one colour per number of people N ; the dotted line marks the 95% target.

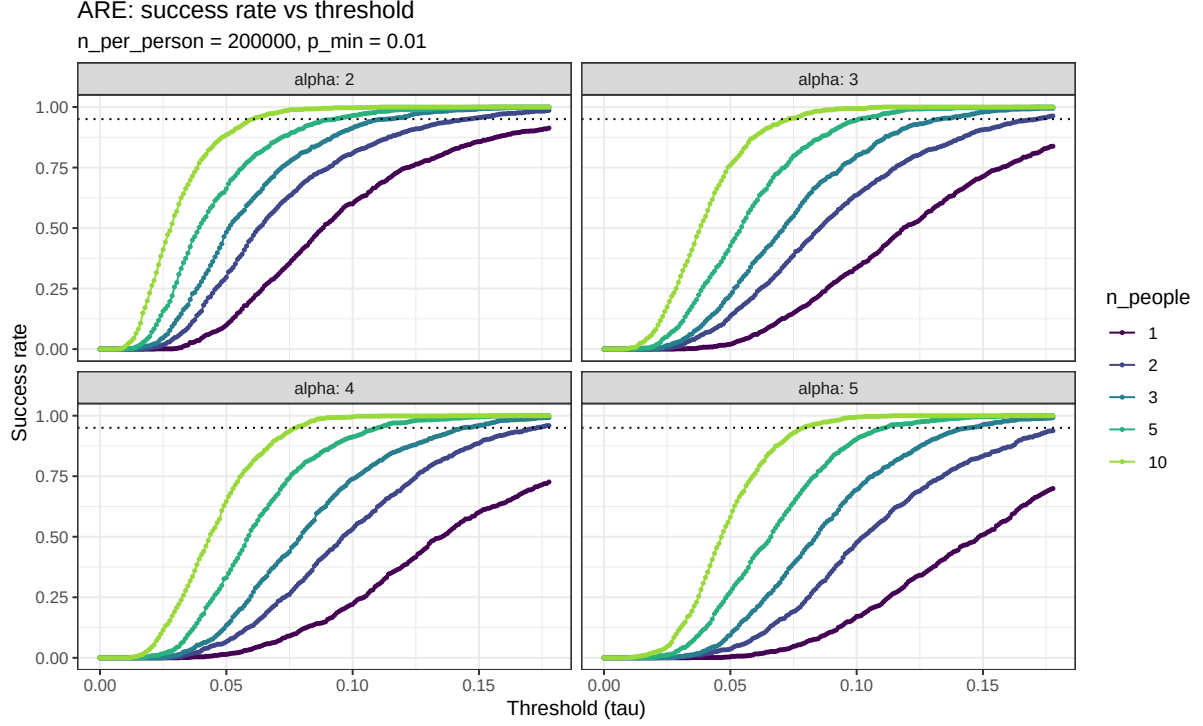


Figure 3: ARE success rate versus threshold τ_{ARE} at 200,000 cells per person.

3.2 Success rate versus cells per person

Figure 4 and Figure 5 fix the threshold ($\tau_{AE} = 0.005$, $\tau_{ARE} = 0.5$) and vary the number of cells per person.

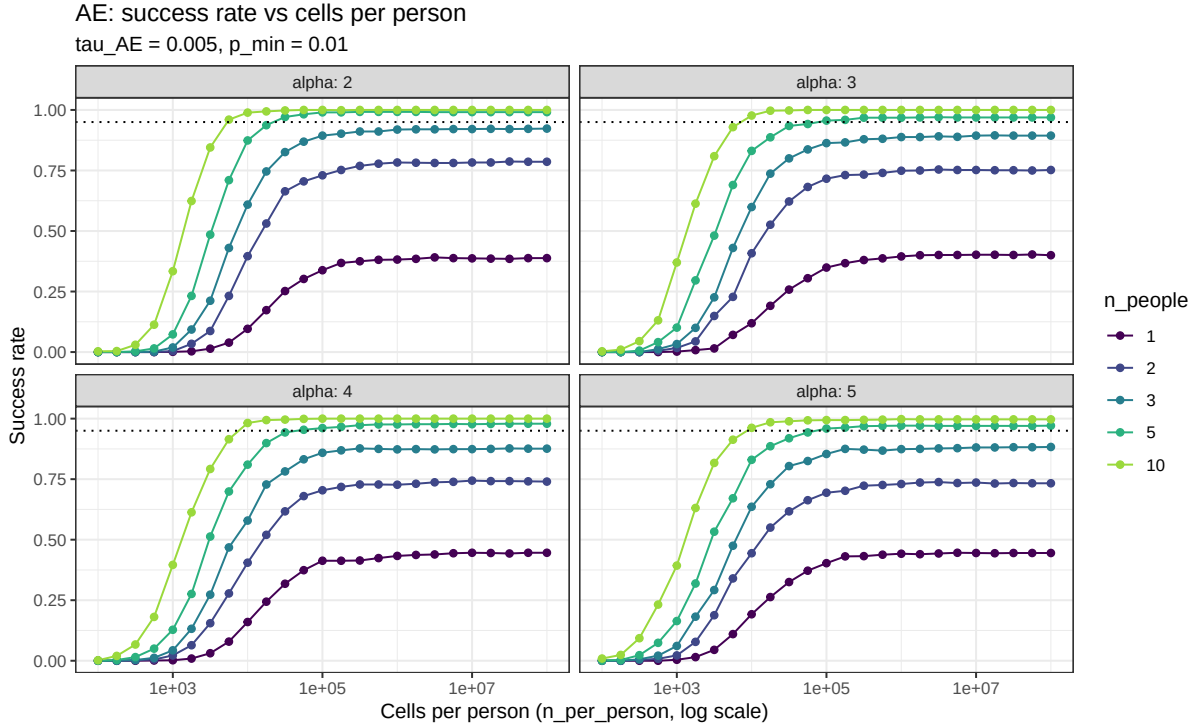


Figure 4: AE success rate versus cells per person (log scale) at $\tau_{AE} = 0.005$. Success rises with cells per person and then reaches a plateau.

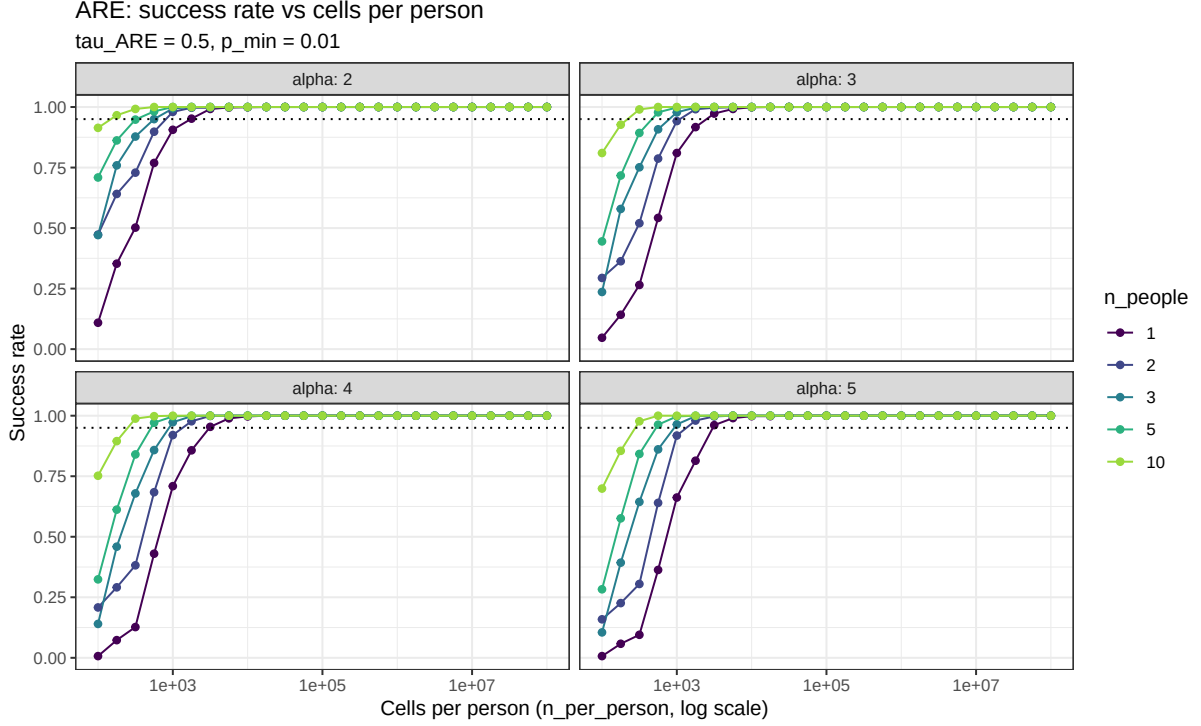


Figure 5: ARE success rate versus cells per person (log scale) at $\tau_{ARE} = 0.5$.

For AE, the success rate increases with the number of cells per person and with the number of people. At $\alpha = 5$ with a single person, it rises from 0% at 10^2 cells to 19% at 10^4 cells, and ends at 44% at 10^8 cells. With 10 people the same curve reaches 96% at 10^4 cells and ends at 100%. The curves flatten, and above roughly 10^6 cells per person extra cells hardly help. The plateau depends mainly on the number of people: with a single person it lies between 39% and 45% across α , and with 10 people it is at least 99.7%.

For ARE the picture is more uniform. From 10^4 cells per person on, the success rate is at least 99.7% for every α and every number of people, so the 95% target is reached everywhere. The differences are at small n . At 10^2 cells with a single person, success is 11% at $\alpha = 2$ and falls to 1% at $\alpha = 5$. With 10 people the values are 91% and 70%. For comparison, without a bound the smallest proportion in the original simulation falls to about $3 \cdot 10^{-6}$ at $\alpha = 5$; here it is never below 0.01.

3.3 Required cells per person

Figure 6 and Figure 7 show, for each α and number of people N , the smallest number of cells per person for which the success rate reaches the 95% target at the fixed thresholds, found with an iterative sample-size solver. Before solving, it is checked whether the target can be reached at all with very many cells per person; combinations where it cannot are marked with $\times$ at the top of the panel.

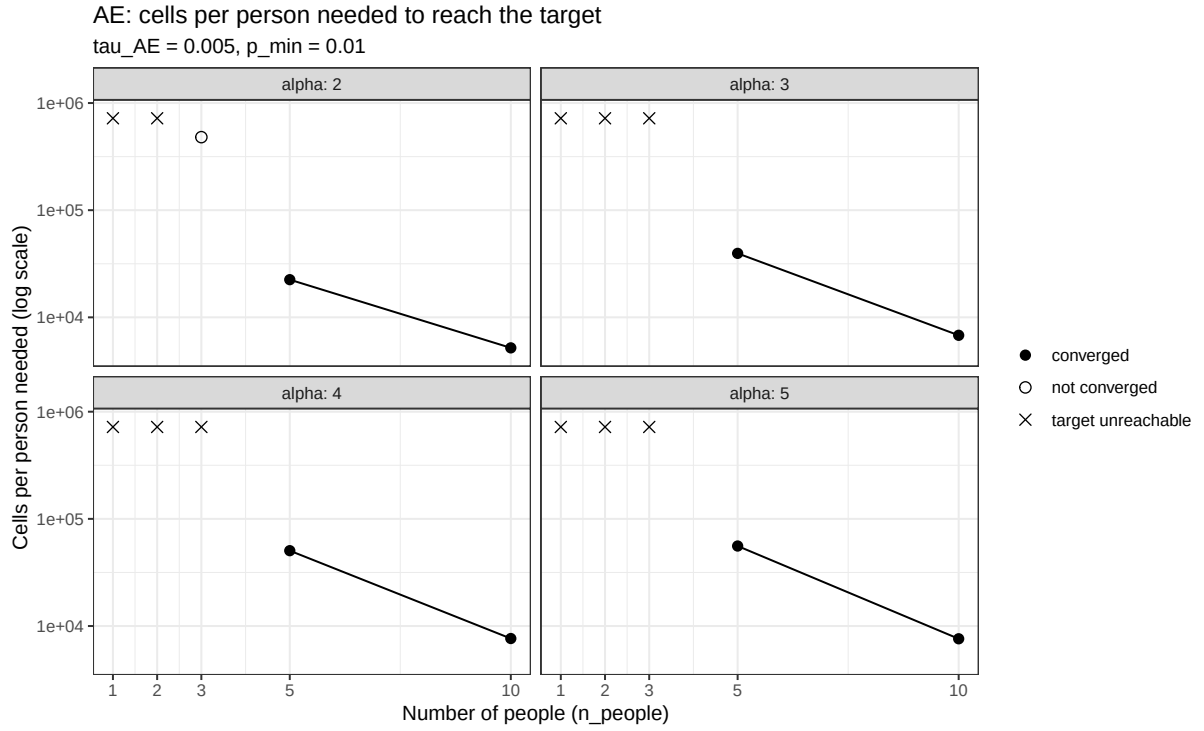


Figure 6: Cells per person needed to reach the 95% AE success target at $\tau_{AE} = 0.005$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.

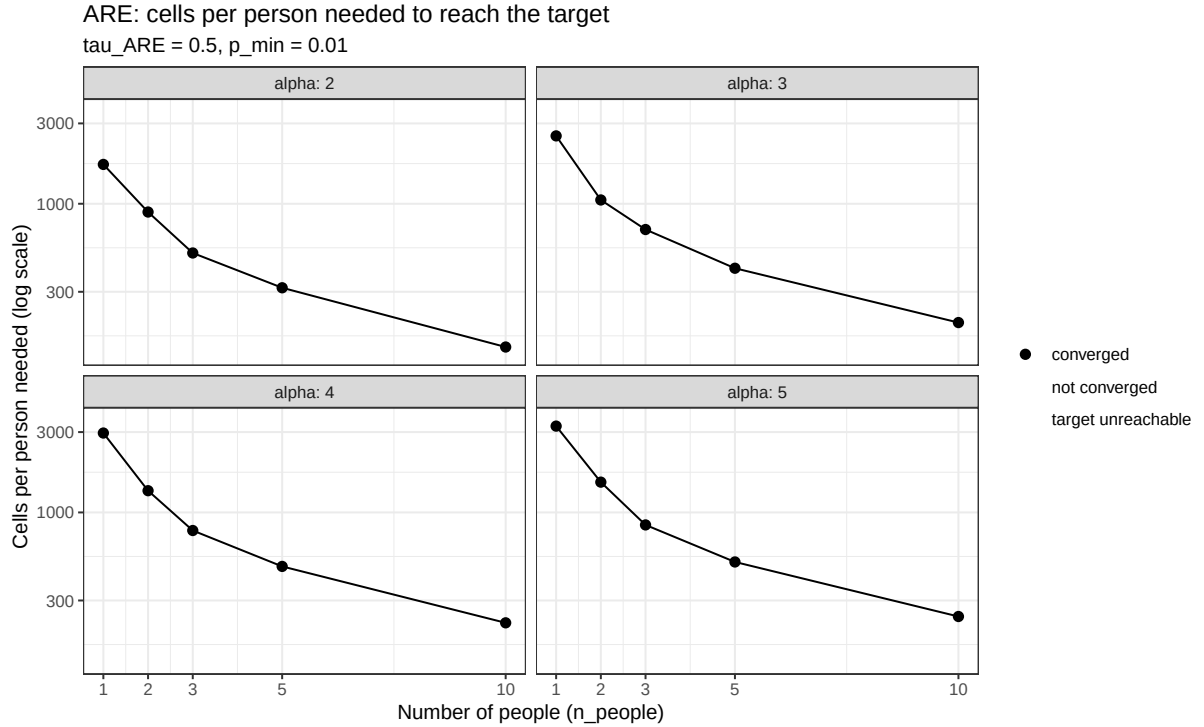


Figure 7: Cells per person needed to reach the 95% ARE success target at $\tau_{ARE} = 0.5$ (log scale), versus the number of people N . One panel per α . An open marker means the solver did not converge; $\times$ at the top means the target cannot be reached.